Q1.

Victoria and Albert play a zero-sum game. The game is represented by the following pay-off matrix for Victoria.

		Albert		
	Strategy	***X***	***Y***	***Z***
Victoria	***P***	3	–1	1
	Q	–2	0	1
	R	4	–1	–1

(a) Find the play-safe strategies for each player.

(3)

(b) State, with a reason, the strategy that Albert should never play.

(1)

(c) (i) Determine an optimal mixed strategy for Victoria.

(5)

(ii) Find the value of the game for Victoria.

(1)

(iii) State an assumption that must be made in order that your answer for part **(c)(ii)** is the maximum expected pay-off that Victoria can achieve.

(1)

(Total 11 marks)

Q2.

John and Danielle play a zero-sum game which does not have a stable solution.

The game is represented by the following pay-off matrix for John.

		Danielle		
	Strategy	***X***	***Y***	***Z***
John	***A***	2	1	–1
	B	–3	–2	2
	C	–3	–4	1

Find the optimal mixed strategy for John.

(Total 6 marks)

Q3.

Kate and Pippa play a zero-sum game. The game is represented by the following pay-off matrix for Kate.

		Pippa		
	Strategy	***D***	***E***	***F***
	A	−2	0	3
Kate	***B***	3	−2	−2
	C	4	1	−1

(a) Explain why Kate should not adopt strategy *B*.

(1)

(b) Find the optimal mixed strategy for Kate and find the value of the game.

(7)

(c) Find the optimal mixed strategy for Pippa.

(4)

(Total 12 marks)

Q4.

Romeo and Juliet play a zero-sum game. The game is represented by the following pay-off matrix for Romeo.

		Juliet		
	Strategy	**D**	**E**	**F**
	A	4	−4	0
Romeo	**B**	−2	−5	3
	C	2	1	−2

(a) Find the play-safe strategy for each player.

(3)

(b) Show that there is no stable solution.

(1)

(c) Explain why Juliet should never play strategy D.

(1)

(d) (i) Explain why the following is a suitable pay-off matrix **for Juliet**.

4	5	−1
0	−3	2

(2)

(ii) Hence find the optimal strategy for Juliet.

(7)

(iii) Find the value of the game for Juliet.

(1)
(Total 15 marks)

Q5.

Two people, Roz and Colum, play a zero-sum game. The game is represented by the following pay-off matrix for Roz.

		Colum		
	Strategy	**C_1**	**C_2**	**C_3**
	R_1	−2	−6	−1
Roz	**R_2**	−5	2	−6
	R_3	−3	3	−4

(a) Explain what is meant by the term ‘zero-sum game’.

(2)

(b) Determine the play-safe strategy for Colum, giving a reason for your answer.

(2)

(c) (i) Show that the matrix can be reduced to a 2 by 3 matrix, giving the reason for deleting one of the rows.

(2)

(ii) Hence find the optimal mixed strategy for Roz.

(7)
(Total 13 marks)

Q6.

(a) Two people, Adam and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Adam.

		Bill		
	Strategy	**B_1**	**B_2**	**B_3**
	A_1	−6	−1	−5
Adam	**A_2**	5	2	−3
	A_3	−5	4	−4

A_4	2	1	−4

(i) Show that this game has a stable solution.

(3)

(ii) Find the play-safe strategy for each player.

(1)

(iii) State the value of the game for **Bill**.

(1)

(b) Roza plays a different zero-sum game against a computer. The game is represented by the following pay-off matrix for Roza.

Computer

	Strategy	**C_1**	**C_2**	**C_3**
	R_1	3	4	−3
Roza	**R_2**	−2	−1	5

(i) State which strategy the computer should never play, giving a reason for your answer.

(1)

(ii) Roza chooses strategy R_1 with probability p. Find expressions for the expected gains for Roza when the computer chooses each of its two remaining strategies.

(2)

(iii) Hence find the value of p for which Roza will maximise her expected gains.

(2)

(iv) Find the value of the game for Roza.

(1)

(Total 11 marks)

Q7.

Two people, Rhona and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rhona.

Colleen

	Strategy	**C_1**	**C_2**	**C_3**
	R_1	2	6	4
Rhona	**R_2**	3	−3	−1
	R_3	x	$x + 3$	3

It is given that $x < 2$.

(a) (i) Write down the three row minima.

(1)

(ii) Show that there is no stable solution.

(3)

(b) Explain why Rhona should never play strategy R_3.

(1)

(c) (i) Find the optimal mixed strategy for Rhona.

(7)

(ii) Find the value of the game.

(1)

(Total 13 marks)

Q8.

Two people, Rowena and Colin, play a zero-sum game.

The game is represented by the following pay-off matrix for Rowena.

Colin

	Strategy	**C_1**	**C_2**	**C_3**
	R_1	–4	5	4
Rowena	**R_2**	2	–3	–1
	R_3	–5	4	3

(a) Explain what is meant by the term ‘zero-sum game’.

(1)

(b) Determine the play-safe strategy for Colin, giving a reason for your answer.

(2)

(c) Explain why Rowena should never play strategy R_3.

(1)

(d) Find the optimal mixed strategy for Rowena.

(7)

(Total 11 marks)

Q9.

Two people, Rob and Con, play a zero-sum game.

The game is represented by the following pay-off matrix for Rob.

Con

Strategy	**C_1**	**C_2**	**C_3**
R_1	–2	5	3

Rob	**R₂**	3	–3	–1
	R₃	–3	3	2

(a) Explain what is meant by the term 'zero-sum game'.

(1)

(b) Show that this game has no stable solution.

(3)

(c) Explain why Rob should never play strategy R_3.

(1)

(d) (i) Find the optimal mixed strategy for Rob.

(7)

(ii) Find the value of the game.

(1)

(Total 13 marks)

Q10.

Sam is playing a computer game in which he is trying to drive a car in different road conditions. He chooses a car and the computer decides the road conditions. The points scored by Sam are shown in the table.

		Road Conditions		
		C_1	C_2	C_3
Sam's Car	S_1	–2	2	4
	S_2	2	4	5
	S_3	5	1	2

Sam is trying to maximise his total points and the computer is trying to stop him.

(a) Explain why Sam should never choose S_1 and why the computer should not choose C_3.

(2)

(b) Find the play-safe strategies for the reduced 2 by 2 game for Sam and the computer, and hence show that this game does not have a stable solution.

(4)

(c) Sam uses random numbers to choose S_2 with probability p and S_3 with probability $1 - p$.

(i) Find expressions for the expected gain for Sam when the computer chooses each of its two remaining strategies.

(3)

(ii) Calculate the value of p for Sam to maximise his total points.

(2)

(iii) Hence find the expected points gain for Sam.

(1)
(Total 12 marks)

Q11.

Two people, Rowan and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rowan.

		Colleen		
	Strategy	**C_1**	**C_2**	**C_3**
Rowan	**R_1**	–3	–4	1
	R_2	1	5	–1
	R_3	–2	–3	4

(a) Explain the meaning of the term 'zero-sum game'.

(1)

(b) Show that this game has no stable solution.

(3)

(c) Explain why Rowan should never play strategy R_1.

(1)

(d) (i) Find the optimal mixed strategy for Rowan.

(7)

(ii) Find the value of the game.

(1)
(Total 13 marks)

Q12.

Two people, Pat and Quigley, play a zero sum game. The game is represented by the following pay-off matrix for Pat.

		Quigley		
	Strategy	**Q_1**	**Q_2**	**Q_3**
Pat	**P_1**	4	3	5
	P_2	–1	5	–2
	P_3	1	2	3

(a) Show that there is no stable solution.

(3)

(b) Explain why Pat should never play strategy P_3.

(1)

(c) Find the optimal mixed strategy for Pat.

(7)

(Total 11 marks)

Q13.

Two people, A and B, play a zero-sum game. The game is represented by the following pay-off matrix for A.

		B		
	Strategy	**I**	**II**	**III**
	I	4	–1	2
A	**II**	2	3	1
	III	1	–2	1

(a) Explain why it will never be optimal for A to adopt strategy III.

(1)

(b) (i) Find the optimal mixed strategy for A.

(7)

(ii) Show that the value of the game is $\frac{7}{5}$.

(1)

(c) By considering optimal mixed strategies for B, show that player B should play strategy II with a probability of $\frac{1}{5}$.

(4)

(Total 13 marks)

Q14.

Arnie (A) and Ben (B) play a zero-sum game. The game is represented by the following pay-off matrix for Arnie (A).

Ben (B)

	I	II	III
I	1	2	2

Arnie (A)	II	3	1	7
	III	2	3	6

(a) Explain why the matrix can be reduced to

3	1
2	3

(2)

(b) Find the optimal mixed strategy for each player and the value of the game.

(9)

(Total 11 marks)

Q15.

Two people, Ann and Bev, play a zero sum game. The game is represented by the following pay-off matrix for Ann.

		Bev		
	Strategy	**I**	**II**	**III**
	I	1	3	6
Ann	**II**	6	1	1
	III	7	4	2

(a) Show that there is no stable solution.

(3)

(b) Explain why it will never be optimal for Ann to adopt **Strategy II** and write down the reduced pay-off matrix for Ann.

(2)

(c) Hence, given that the value of the game is $3\frac{3}{5}$, find the optimal mixed strategy for **Bev**.

(*You are **not** required to find the optimal mixed strategy for Ann*.)

(5)

(Total 10 marks)

Q16.

Asif and Bharveen play a zero-sum game. The game is represented by the following pay-off matrix for Asif.

Bharveen

		I	II	III
Asif	I	4	2	6
	II	2	10	1
	III	3	1	5

(a) Explain why Asif should never play strategy **III**.

(1)

(b) Asif plays strategy **I** with probability p_1 and strategy **II** with probability p_2. The value of the game is v. Use the matrix to write down **three** inequalities in the form $v \leq ap_1 + bp_2$.

(3)

The probabilities p_1 and p_2 also satisfy the inequalities

$$p_1 + p_2 \leq 1, \quad p_1 \geq 0 \quad \text{and} \quad p_2 \geq 0.$$

The Simplex tableau for the linear programming problem which will give the optimal strategy for Asif can be written as:

P	v	p_1	p_2	r	s	t	u	
1	–1	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0	1
0	1	–4	–2	0	1	0	0	0
0	1	–2	–10	0	0	1	0	0
0	1	–6	–1	0	0	0	1	0

(c) (i) Apply **one** iteration to this Simplex tableau.

(4)

(ii) Explain why your answer to part (c)(i) is **not** optimal.

(1)

(iii) After two further iterations, Asif s tableau is:

P	v	p_1	p_2	r	s	t	u	
5	0	0	0	18	4	1	0	18
0	0	0	10	2	1	–1	0	2
0	0	10	0	8	–1	1	0	8
0	5	0	0	18	4	1	0	18

0	0	0	0	14	–13	3	10	14

Show that the value of the game in this case is $3\frac{3}{5}$ and find Asif's optimal mixed strategy.

(3)

(d) Assuming the value of the game is $3\frac{3}{5}$ and by considering mixed strategies, find the optimal mixed strategy for Bharveen.

(6)

(Total 18 marks)

Q17.

Alan and Tracy play squash against each other every week. They each have three shots that they can play to try to win a point: a boast (B), a drop shot (D) and a lob (L). The pay-off matrix for Alan is given below.

Tracy

		B	D	L
Alan	B	12	5	2
	D	5	–2	–3
	L	8	10	13

(a) State which shot Alan should **not** play, giving a reason for your answer.

(2)

(b) By considering mixed strategies, find the ratio of the number of times Alan should play each of the other two shots.

(8)

(c) Find the value of the game.

(1)

(Total 11 marks)

Q18.

Two people, A and B, play a zero sum game. The game is represented by the following pay-off matrix for A.

B

	Strategy	I	II	III
	I	5	1	3
A	II	2	5	4

III	4	−1	2

(a) Show that there is no stable solution.

(2)

(b) Explain why it will never be optimal for A to adopt strategy III.

(1)

(c) By considering mixed strategies, and giving your answers as exact fractions,

(i) find the optimal mixed strategy for A,

(7)

(ii) find the value of the game.

(1)

(Total 11 marks)

Q19.

Alan (A) and Betty (B) play a zero-sum game. The game is represented by the following pay-off matrix for Alan (A).

		B I	B II	B III
	I	5	2	7
A	II	4	1	5
	III	3	5	6

(a) Explain why this matrix can be reduced to

5	2
3	5

(3)

(b) Find the optimal mixed strategy for each player and the value of the game.

(9)

(Total 12 marks)